

ECE 3204 Microelectronic Circuits II

Lab #7: INTEGRATING A/D CONVERTER

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AK 227

# Table of Contents

|                     |       |                |
|---------------------|-------|----------------|
| <b>Abstract</b>     | ----- | <b>pg. 3</b>   |
| <b>Equipment</b>    | ----- | <b>pg. 3</b>   |
| <b>Introduction</b> | ----- | <b>pg. 3</b>   |
| <b>W1</b>           | ----- | <b>pg. 4-8</b> |
| <b>Discussion</b>   | ----- | <b>pg. 9</b>   |
| <b>Conclusion</b>   | ----- | <b>pg. 9</b>   |

## Abstract

In this lab, we investigated the use of an integrating analog-to-digital converter (ADC) for converting an analog input voltage into a time-based digital measurement. The ADC was built using an LMC6484 op-amp configured as an integrator, with analog switches controlling the reset, input integration, and reference integration phases. During operation, the integrator was first reset to zero, then used to integrate the input voltage for a fixed time interval. The circuit then switched to a negative reference voltage, causing the integrator output to ramp back toward zero. An LM311 comparator was used to detect when the integrator output crossed zero, producing an output pulse whose width depended on the input voltage. By measuring the pulse width of  $V_{pulse}$ , the input voltage could be determined based on the dual-slope ADC relationship. The input-output transfer characteristic was then measured across the ADC range to evaluate linearity and observe possible nonideal behavior caused by capacitor effects. This demonstrated how integrating ADCs can accurately convert analog voltage levels into measurable timing information.

## EQUIPMENT

- ECE Lab Kit
- LM311 comparator
- 74HC4053 triple 2:1 analog switch
- DC Power Supply
- Digital Multimeter (DMM)
- Function Generator
- Oscilloscope

## Introduction

For this lab, sampled data circuits were used to design and analyze an integrating analog-to-digital converter (ADC). The ADC was implemented using an LMC6484 op-amp as an integrator, with analog switches controlling the reset, input integration, and reference integration stages. The input voltage was integrated for a fixed time, and then a negative reference voltage was applied to ramp the integrator output back toward zero. An LM311 comparator was then used to detect the zero crossing and produce an output pulse, where the pulse width is dependent on the input voltage. This demonstrated how an analog voltage can be converted into a measurable time interval using dual-slope ADC operation.

## W1 - Results:

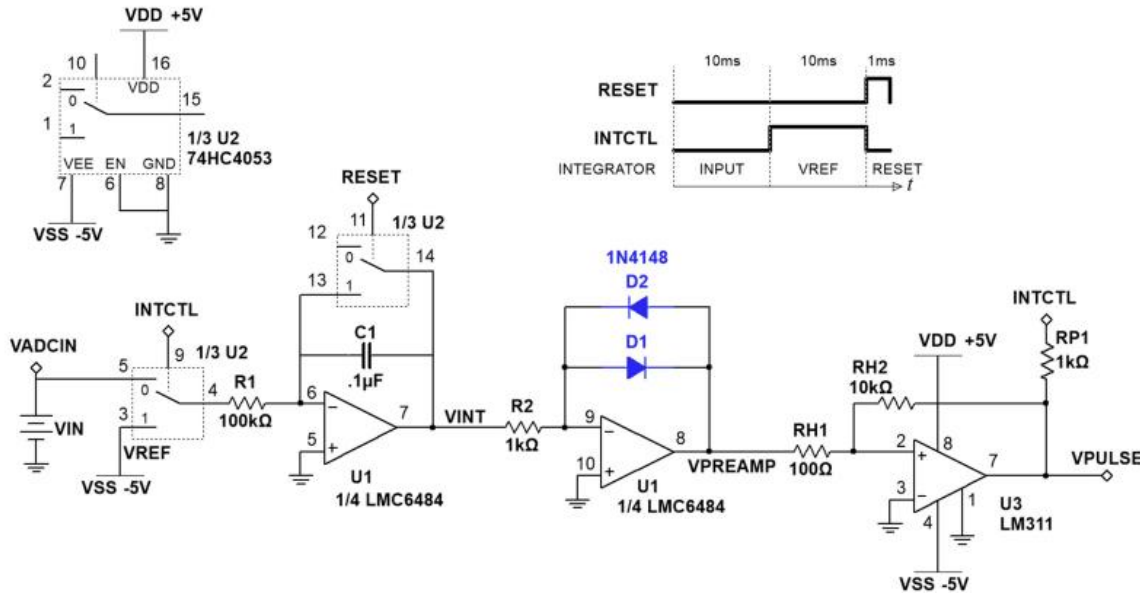


Figure 1.1 Schematic for integrating ADC build.

### L1 Circuit Description

This circuit uses an ADC in 4 main blocks with an input switching stage, integrator stage, a preamplifier, and comparator output stage. The input stage uses the 74HC4053 analog switch that allows it to select between the  $V_{ADCIN}$  input voltage and a known reference voltage of -5V and the selection is controlled by INTCTL signal. The second switch controlled by RESET, is placed across the integrating capacitor  $C_1$  to be able to discharge it and force integrator output to 0V at the beginning of each conversion cycle. The inverting integrator is built using the LMC6484 op-amp, resistor  $R_1$  of 100kΩ, and capacitor  $C_1$  of 0.1μF. When input voltage is selected, the integrator produces an output of a negating slope ramp with slope proportional to input voltage. When reference voltage is selected, it produces a positive slope ramp with a constant slope. The integrated output is then given to the preamplifier stage that had two diodes clamp the preamplifier output to approximately  $\pm 0.7V$ , preventing large voltage swing and avoid slew rate limitations. Lastly, the signal enters the LM311 comparator which detects when the signal crosses 0V. The comparator output is pulled up through the resistor this is tied to the INTCTL signal which effectively creates an AND function so that the output of  $V_{PULSE}$  is only occurring in the reference integration phase.

W1-1

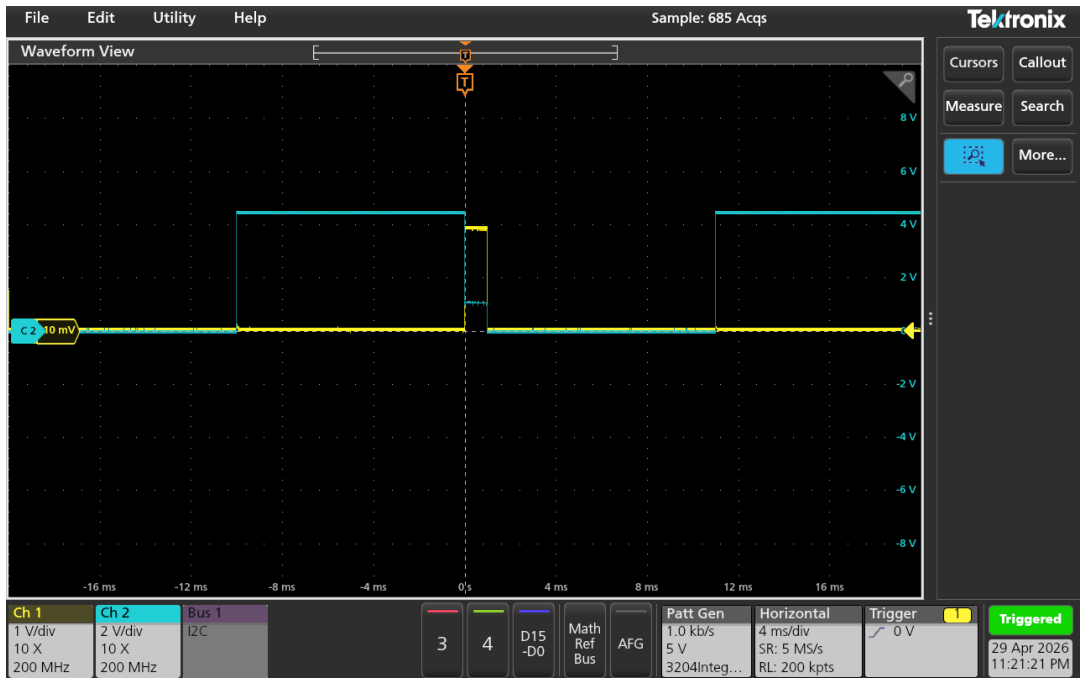


Figure 1-2: L1-1 waveform

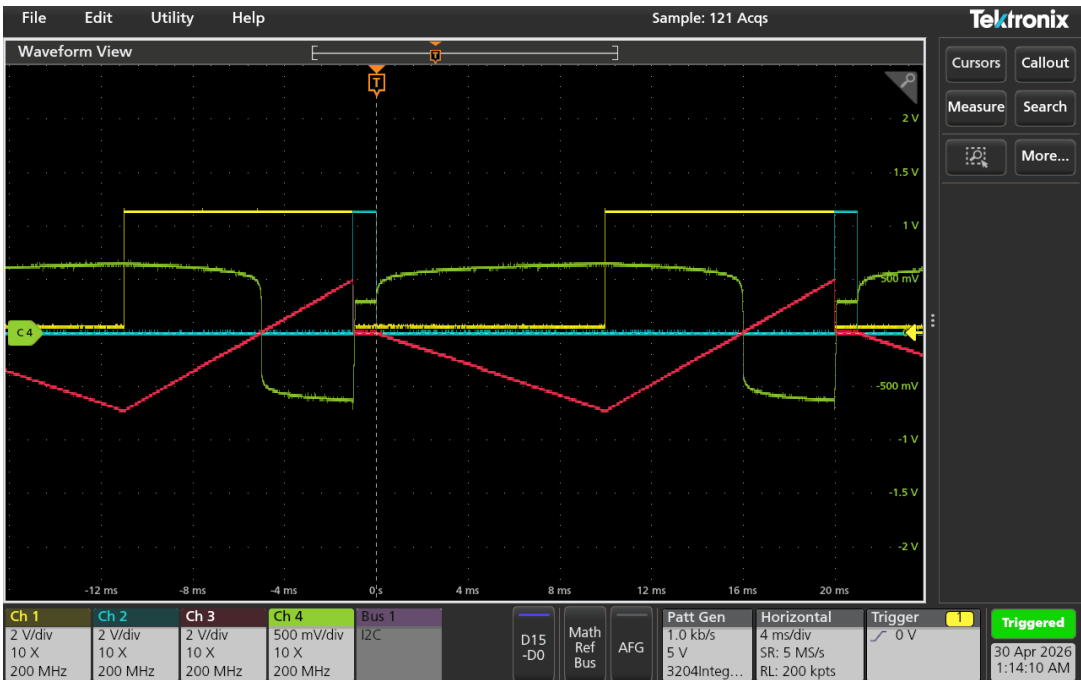


Figure 1-3: L1-3 waveform

The waveforms from L1-1 and L1-3 confirm the correct timing and operations if integration from the ADC. The timing waveform, RESET was seen to go high for about 1ms which discharges the capacitor and forces the integrated output of  $V_{INT}$  to become 0V. After this the INTCTL signal selects an input voltage from about 10ms which is then followed by switching to a reference voltage for an additional 10ms. This confirms that the required time sequence of going from reset to input to integration to reference integration is working correctly. From the integrator waveform of  $V_{INT}$  it is seen to ramp negatively in the integration phase with its slope being dependent on the magnitude of  $V_{ADCIN}$ . When the circuit changes to the reference voltage, the waveform then ramps positively at a constant slope till it reaches about 0V. This accurately matches the behavior of a dual slop ADC operation with the negative ramp encoding the input voltage and positive ramp that determines the time needed to return to 0V.

## W1-2

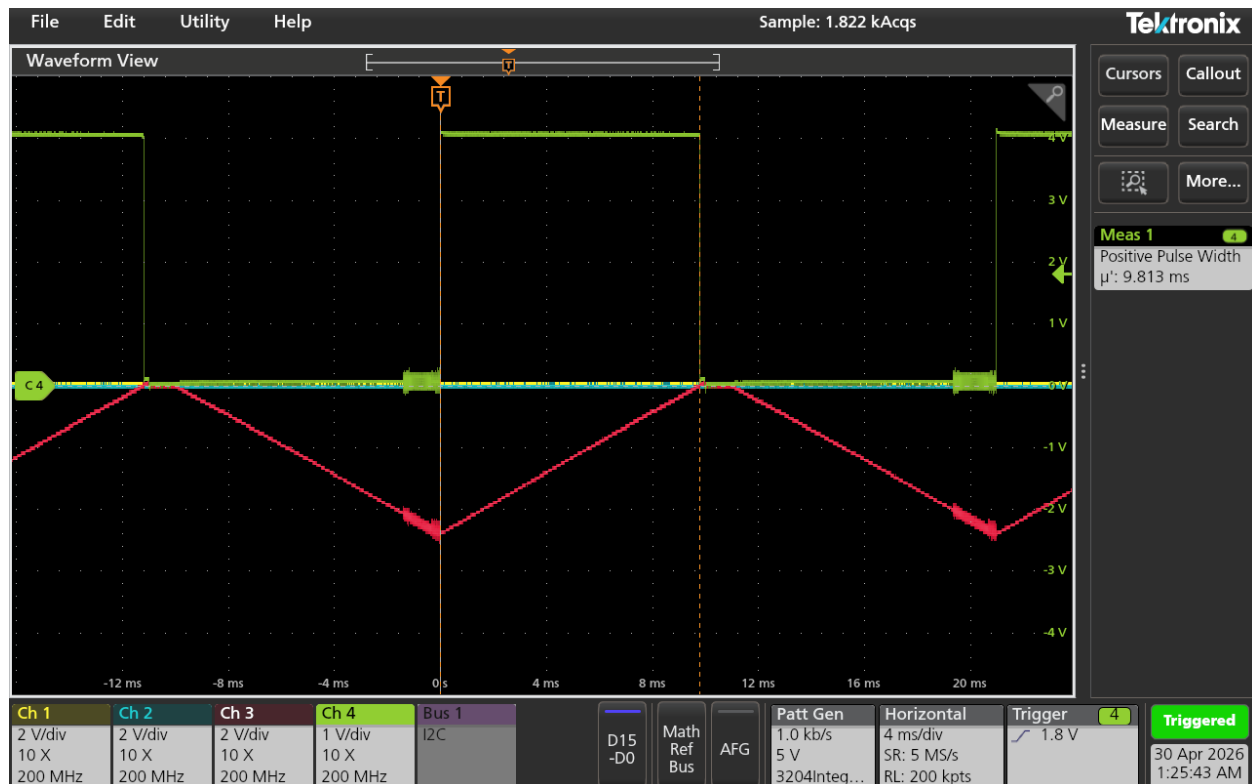


Figure 1-4: L1-6 waveform

| VADCIN (V) | Pulse Width (ms) | Pulse Width IDEAL (ms) | Error (ms) |
|------------|------------------|------------------------|------------|
| 0.09994    | 0.1978           | 0.19988                | -0.00208   |
| 0.49995    | 0.9855           | 0.9999                 | -0.0144    |
| 0.8995     | 1.799            | 1.799                  | 0          |
| 1.2996     | 2.601            | 2.5992                 | 0.0018     |
| 1.6995     | 3.403            | 3.399                  | 0.004      |
| 2.0995     | 4.204            | 4.199                  | 0.005      |
| 2.4991     | 5.006            | 4.9982                 | 0.0078     |
| 2.899      | 5.808            | 5.798                  | 0.01       |
| 3.2989     | 6.61             | 6.5978                 | 0.0122     |
| 3.6986     | 7.412            | 7.3972                 | 0.0148     |
| 4.0984     | 8.213            | 8.1968                 | 0.0162     |
| 4.4983     | 9.014            | 8.9966                 | 0.0174     |
| 4.8981     | 9.811            | 9.7962                 | 0.0148     |

Figure 1-5: Table of results from L1-6 and ideal values

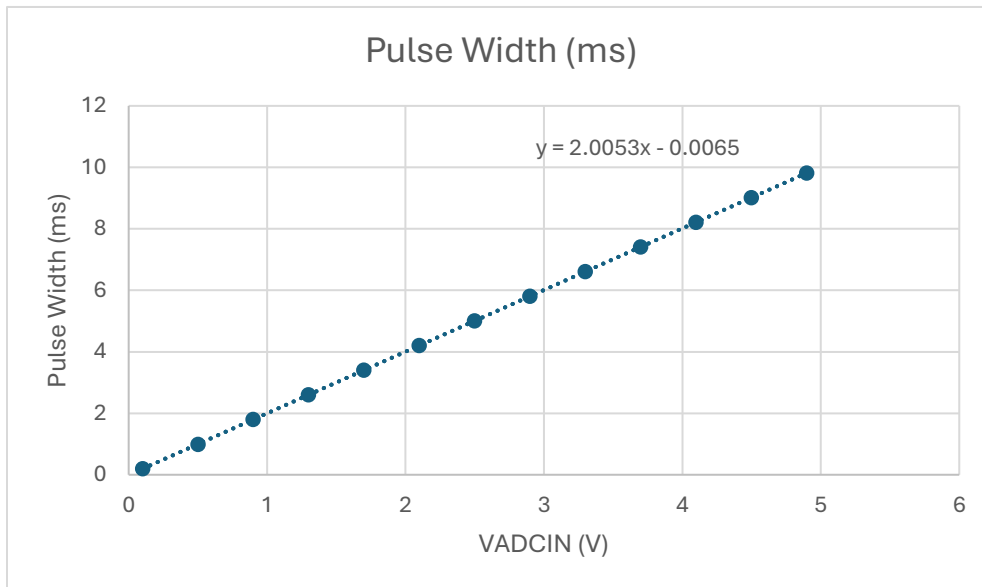


Figure 1-6: Plot showing the input-output characteristic from input DC voltage to measured duration of the VPULSE signal at the comparator output

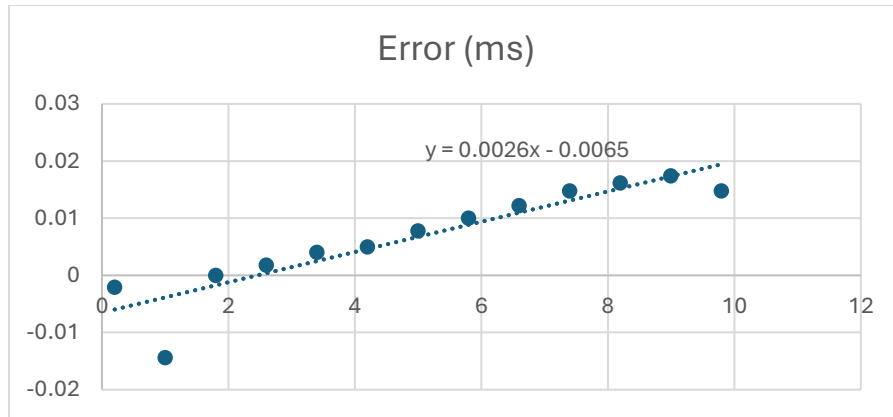


Figure 1-7: Error Plot of Figure 1-x

The measured input – output data shown above shows a strong linear relationship between the input voltage of  $V_{ADCIN}$  and the pulse width  $V_{PULSE}$  which would be expected for a dual slope ADC. This relationship is shown in the equation of  $T_{PULSE} = V_{IN}/V_{REF} * T_{INT} = 0.002 * V_{IN}$  in seconds with  $V_{REF} = 5V$  and  $T_{INT} = 10ms$ . The measured pulse widths are very close to the ideal value of 2ms/V in this case at 2.0053ms/V based off the trendline in Figure 1-6. The error plot shows the small deviations between the ideal linear response and has error increasing slightly as the input voltage increases. With error ranging from -0.002ms at the low voltages to +0.017ms at high voltages. If 1LSB is 1 $\mu$ s, the peak linearity would be 0.017ms/1 $\mu$ s = 17LSB. This shows that the ADC has pretty strong linearity with some minor systematic errors and is consistent with the expected non ideal parts of a practical ADC integrating circuit.



## Discussion

During this lab, sampled data systems were analyzed using an integrating analog-to-digital converter. In W1, the ADC timing and integrator operation were successfully verified. The RESET signal went high for about 1 ms, discharging the capacitor and forcing the integrator output back to 0V before each conversion cycle. The INTCTL signal then selected the input voltage for about 10 ms, followed by the reference voltage for another 10 ms. This matched the expected timing sequence for reset, input integration, and reference integration. The integrator waveform also showed the expected dual-slope behavior. During the input integration phase,  $V_{INT}$  ramped negatively, with the slope depending on the magnitude of  $V_{ADCIN}$ . When the circuit switched to the reference voltage, the waveform ramped positively at a more constant slope until it returned toward 0 V. This confirmed that the integrator and analog switch stages were functioning properly. In W1-2, the measured input-output characteristic showed a strong linear relationship between  $V_{ADCIN}$  and the pulse width of  $V_{PULSE}$ . The measured trendline had a slope of about 2.0053 ms/V, which was very close to the ideal value of 2 ms/V. The error plot showed small deviations from the ideal response, with the error increasing slightly at higher input voltages. The peak nonlinearity was about 17 LSB, assuming 1 LSB equals 1  $\mu$ s. Overall, the results closely matched theoretical predictions, with minor non-ideal effects likely caused by component tolerances, capacitor nonlinearity, and measurement limitations.

## Conclusion

In conclusion, this lab demonstrated how an integrating analog-to-digital converter can convert an analog input voltage into a measurable time-based output. The ADC used an integrator, analog switches, and a comparator to create a pulse width that changed depending on the value of  $V_{ADCIN}$ . We found that the measured pulse width increased linearly with the input voltage, closely matching the expected dual-slope ADC relationship. Some small deviations were present in the error plot, likely due to non-ideal components, capacitor behavior, and measurement limitations. This lab reinforced concepts such as integration, comparator operation, ADC linearity, and the relationship between analog voltage and digital timing measurements.